

VIKRANT VIJAYBHAI JOLIYA

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Professional Summary

Backend Software Engineer specializing in scalable Spring Boot architectures and AI integration with around 3 years of enterprise experience at TCS. Claude Certified Architect with a proven track record of delivering high-impact, AI solutions for Fortune 500 clients. Expert in full-stack development using Java, Spring Boot, Python, and PostgreSQL. Demonstrated success in executing rapid production deployments under high pressure, modernizing legacy systems, and engineering AI-assisted workflows that significantly reduce processing times and accelerate feature delivery.

Work Experience

Software Developer

11/2023 – Present

Tata Consultancy Services

India

- **AI Cybersecurity Threat Detector (Hackathon Winner):** Led team "HAVD Gurus" to 1st place in the enterprise AI Friday Hackathon by architecting an intelligent security agent. Engineered a Python and AWS-based pipeline that dynamically analyzes complex security logs, detects anomaly patterns, and automatically generates targeted remediation steps, significantly reducing incident response time
- **Rapid Production Deployments & Caching Strategy:** Engineered and successfully deployed a comprehensive caching strategy in the production environment under strict deadlines, drastically optimizing system response times. Conceptualized and launched an end-to-end dashboard workflow on very short notice, demonstrating the ability to deliver mission-critical architectural features under high-pressure constraints
- **Enterprise Scale Spring Boot Architecture:** Spearheaded the modernization of a 100K+ LOC legacy system for GE Aerospace, orchestrating a zero-downtime migration from Spring Boot 1.5 to 3.4 and Java 8 to 17. Redesigned authentication using Spring Security 6.x and optimized Hibernate ORM, achieving a 30% performance boost and eliminating 50+ critical CVEs
- **AI-Driven Development & Prototyping:** Accelerated feature delivery and test generation by adopting AI-assisted coding workflows utilizing GitHub Copilot and LLM agents. Leveraged tools like LangChain to rapidly prototype and integrate generative AI capabilities within strict, air-gapped enterprise security environments
- **AWS Serverless & Data Engineering:** Architected high-performance AWS serverless pipelines and optimized Data Lake scheduling algorithms for manufacturing data. Slashed processing time for 60,000+ critical records from 2 months to just 15 days (75% reduction) using parallel processing, custom PostgreSQL functions, and intelligent caching
- **Database Engineering Excellence:** Designed custom PostgreSQL functions, stored procedures, and triggers for high-volume OLTP operations; optimized Hibernate 6.x configurations including lazy loading strategies, batch processing, and HikariCP connection pooling—delivering 40% performance improvement across large-scale transactions
- **Security Hardening & Compliance:** Performed comprehensive security audits on legacy codebases, identifying and remediating 25+ critical CVEs using OWASP best practices; upgraded authentication mechanisms with enhanced token validation—achieving 100% compliance with CMMC Level 2/3 policies
- **SPA Integration & Modern Web Architecture:** Designed sophisticated Single Page Application routing solution for Spring Boot backend; implemented intelligent resource resolver with fallback mechanisms and multi-environment CORS policies—eliminating 404 routing errors and enabling seamless React/Angular integration

Education

Kalasalingam Academy of Research and Education

2019 – 2023

Bachelor of Technology in Computer Science & Engineering – First Class with Distinction

CGPA: 9.42/10.0

Technical Skills

Programming Languages: Java, Python, SQL, PL/pgSQL, Shell Scripting
Frameworks & Libraries: Spring Boot, JPA, Hibernate, FastAPI, Flask, LangChain
Databases & Storage: PostgreSQL, AWS S3, Database Design, Query Optimization, Stored Procedures and CTE
Cloud & DevOps: AWS (S3, Lambda, EC2, Comprehend, Translate, Polly), Docker, Jenkins, CI/CD Pipelines, Git
APIs & Integration: REST API Development, Microservices Architecture, JSON/XML Processing, Swagger/OpenAPI
Methodologies: Agile/Scrum, Test-Driven Development (TDD), AI-Assisted Development, System Design

Research Publications & Academic Contributions

IEEE CISES 2023 Conference: "Honeypot Implementation for Enhanced Network Security using Docker and AWS"

Published research on containerized cybersecurity infrastructure, demonstrating cloud-native security monitoring and threat detection systems

IEEE ICOEI 2022 Conference: "SOS Android Application for Emergency Response"

Developed and published research on location-based emergency response mobile application with real-time GPS tracking and automated alert systems

Certifications & Awards

Claude Certified Architect (Foundations): Validates foundational expertise in Anthropic's Claude models, generative AI principles, and prompt engineering applications

AWS Certified Developer – Associate (DVA-C02): Amazon Web Services certification validating proficiency in developing, deploying, debugging cloud applications, and implementing security best practices

AI Friday Hackathon Winner (2025): 1st place in company-wide innovation challenge for developing intelligent Log Analyzer Tool with AI-powered anomaly detection capabilities

IBM SkillsBuild – Cyber Security & Forensics Graduate: Comprehensive training in network security, digital forensics, threat analysis, penetration testing, and security best practices